

0.1 Mathematical Background

0.2 Magnetic Field Derivation

Maxwell's equations in free space are linear [?, p. 59] so the total magnetic field will be the sum of each individual straight line. Our main goal then, is to find the magnetic field generated at an observer point p due to the current flowing along $\vec{a}$.

We start by defining vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$.

- $\vec{a}$ points along the current filament.
- $\vec{b}$ points from the observer to the end of the filament.
- $\vec{c}$ points from the observer to the start of the filament.

Note that throughout I use the convention $|\vec{a}| = a$.

Now, we use the Biot Savart Law to find the magnetic field in each of the lines.

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times \vec{r}}{r^3} \quad (1)$$

To work out this integral we first parametrise $\vec{r}$ in time, yielding:

$$\begin{aligned} \vec{r}(t) &= \vec{c} + \vec{a}t, \quad t \in [0, 1] \\ d\vec{l} &= \vec{a}dt \end{aligned}$$

Substituting in gives:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int_0^1 \frac{(\vec{a}dt) \times (\vec{c} + \vec{a}t)}{|\vec{c} + t\vec{a}|^3} \quad (2)$$

Noting that $\vec{a} \times \vec{a} = 0$, and that $\vec{a}$ and $\vec{c}$ are constants, this simplifies to:

$$\vec{B} = \frac{\mu_0 I}{4\pi} (\vec{a} \times \vec{c}) \int_0^1 \frac{dt}{|\vec{c} + t\vec{a}|^3} \quad (3)$$

The denominator must now be simplified.

$$|\vec{c} + t\vec{a}|^3 = (c^2 + t^2 a^2 + 2t(\vec{a} \cdot \vec{c}))^{3/2} \quad (4)$$

For simplicity, let $\beta = (\vec{a} \cdot \vec{c})$

$$|\vec{c} + t\vec{a}|^3 = (a^2t^2 + 2\beta t + c^2)^{3/2} \quad (5)$$

Completing the square:

$$|\vec{c} + t\vec{a}|^3 = a^3 \left(\left(t + \frac{\beta}{a^2} \right)^2 - \frac{\beta^2}{a^4} + \frac{c^2}{a^2} \right)^{3/2} \quad (6)$$

Now let $u = t + \frac{\beta}{a^2}$, $du = dt$, $u(0) = \frac{\beta}{a^2}$, $u(1) = 1 + \frac{\beta}{a^2}$. Also, let $\delta = \frac{c^2}{a^2} - \frac{\beta^2}{a^4}$. Subbing back into our Biot Savart Formula gives:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^3} \int_{\frac{\beta}{a^2}}^{1+\frac{\beta}{a^2}} \frac{du}{(u^2 + \delta)^{3/2}} \quad (7)$$

Want to use the identity $\sec^2 \phi = 1 + \tan^2 \phi$, to do so, multiply across by a factor of $\delta^{3/2}$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^3 \delta^{3/2}} \int_{\frac{\beta}{a^2}}^{1+\frac{\beta}{a^2}} \frac{du}{\left(\frac{u^2}{\delta} + 1\right)^{3/2}} \quad (8)$$

Now that it's in the correct form let: $\tan^2 \phi = \frac{u^2}{\delta}$, then:

$$\begin{aligned} \tan \phi &= \frac{u}{\sqrt{\delta}} \\ \sec^2 \phi d\phi &= \frac{1}{\sqrt{\delta}} du \\ \phi \left(\frac{\beta}{a^2} \right) &= \arctan \left(\frac{\beta}{a^2 \sqrt{\delta}} \right) \\ \phi \left(1 + \frac{\beta}{a^2} \right) &= \arctan \left(\frac{1 + \frac{\beta}{a^2}}{\sqrt{\delta}} \right) \end{aligned}$$

Substituting back in:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^3 \delta^{3/2}} \int_{\arctan\left(\frac{\beta}{a^2 \sqrt{\delta}}\right)}^{\arctan\left(\frac{1}{\sqrt{\delta}}\left(1 + \frac{\beta}{a^2}\right)\right)} \frac{\sqrt{\delta} \sec^2 \phi d\phi}{(1 + \tan^2 \phi)^{3/2}} \quad (9)$$

Using the identities $\sec^2 \phi = 1 + \tan^2 \phi$ and $\cos \phi = \frac{1}{\sec \phi}$ and simplifying yields:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^3 \delta} \int_{\arctan\left(\frac{\beta}{a^2 \sqrt{\delta}}\right)}^{\arctan\left(\frac{1}{\sqrt{\delta}}\left(1 + \frac{\beta}{a^2}\right)\right)} \cos \phi d\phi \quad (10)$$

Solving:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^3 \delta} \left(\sin \left(\arctan \left(\frac{1}{\sqrt{\delta}} \left(1 + \frac{\beta}{a^2} \right) \right) \right) - \sin \left(\arctan \left(\frac{\beta}{a^2 \sqrt{\delta}} \right) \right) \right) \quad (11)$$

Now, we must consider a right angled triangle to work this out. $\sin \phi = \frac{o}{h}$, $\tan \phi = \frac{o}{a}$ therefore $\sin \left(\arctan \left(\frac{o}{a} \right) \right) = \frac{o}{\sqrt{o^2 + a^2}}$ by pythagoras.

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^3 \delta} \left(\left(\frac{1 + \frac{\beta}{a^2}}{\sqrt{\delta + \left(1 + \frac{\beta}{a^2} \right)^2}} \right) - \left(\frac{\beta}{\sqrt{\beta^2 + a^4 \delta}} \right) \right) \quad (12)$$

Recalling $\delta = \frac{c^2}{a^2} - \frac{\beta^2}{a^4}$, $\beta = \vec{a} \cdot \vec{c}$, and looking at each of the denominators:

$$\begin{aligned} \sqrt{\delta + \left(1 + \frac{\beta}{a^2} \right)^2} &= \sqrt{\frac{c^2}{a^2} - \frac{\beta^2}{a^4} + 1 + 2\frac{\beta}{a^2} + \frac{\beta^2}{a^4}} \\ &= \sqrt{1 + \frac{c^2}{a^2} + \frac{2}{a^2} (\vec{a} \cdot \vec{c})} \\ &= \frac{1}{a} \sqrt{a^2 + c^2 + 2(\vec{a} \cdot \vec{c})} \\ &= \frac{1}{a} |\vec{a} + \vec{c}| \\ &= \frac{b}{a} \end{aligned}$$

$$\begin{aligned} \sqrt{\beta^2 + a^4 \delta} &= \sqrt{\beta^2 + a^2 c^2 - \beta^2} \\ &= \sqrt{a^2 c^2} \\ &= ac \end{aligned}$$

Subbing into our formula for $\vec{B}$ now gives:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{ac^2 - \frac{(\vec{a} \cdot \vec{c})^2}{a}} \left(\frac{1 + \frac{\vec{a} \cdot \vec{c}}{a^2}}{\frac{b}{a}} - \frac{\vec{a} \cdot \vec{c}}{ac} \right) \quad (13)$$

Simplifying a small bit gives:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{a^2 c^2 - (\vec{a} \cdot \vec{c})^2} \left(\frac{a^2 + \vec{a} \cdot \vec{c}}{b} - \frac{\vec{a} \cdot \vec{c}}{c} \right) \quad (14)$$

Now, picking out these two parts and simplifying, where θ is the angle subtended by $\vec{a}$ and $\vec{c}$:

$$\begin{aligned} a^2 c^2 - (\vec{a} \cdot \vec{c})^2 &= a^2 c^2 - a^2 c^2 \cos^2(\theta) \\ &= a^2 c^2 (1 - \cos^2 \theta) \\ &= a^2 c^2 \sin^2 \theta \\ &= |\vec{a} \times \vec{c}|^2 \end{aligned}$$

$$\begin{aligned} a^2 + \vec{a} \cdot \vec{c} &= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{c} \\ &= \vec{a} \cdot (\vec{a} + \vec{c}) \\ &= \vec{a} \cdot \vec{b} \end{aligned}$$

This gives us our final form:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{a} \times \vec{c}}{|\vec{a} \times \vec{c}|^2} \left(\frac{\vec{a} \cdot \vec{b}}{b} - \frac{\vec{a} \cdot \vec{c}}{c} \right) \quad (15)$$

Now, remembering that this equation is linear and that each coil is made up of straight line filaments we get:

$$\vec{B}_{\text{total}} = \sum_{\text{coils}} \sum_{\text{filaments}} \vec{B}_{\text{filament}} \quad (16)$$

Now, since we are in free space we use the identity $\vec{H} = \frac{\vec{B}}{\mu_0}$ [?, p. 285]

0.3 Understanding the forward map.

The problem is framed as an inverse problem. The forward map takes position and orientation data and calculates 8 flux values from each of the coils. Our goal is to approximate an inverse to this map. To do so, we first try to understand the forward map

0.3.1 Definition and properties.

First we define a pose vector $\mathbf{p} \in \mathbb{R}^5$

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \quad (17)$$

In actuality, $\mathbf{p}$ is restricted to a set $\Omega \subset \mathbb{R}^5$

$$\Omega = \left\{ \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \in \mathbb{R}^5 : \begin{array}{l} x \in (-0.125, 0.125) \\ y \in (-0.125, 0.125) \\ z \in (0, 0.25) \\ \theta \in (0, \pi) \\ \phi \in [0, 2\pi) \end{array} \right\} \quad (18)$$

Then we define a vector of fluxes, $\Phi \in \mathbb{R}^8$

$$\Phi = \begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \\ \Phi_4 \\ \Phi_5 \\ \Phi_6 \\ \Phi_7 \\ \Phi_8 \end{bmatrix} \quad (19)$$

Now we can write the forward map: $\Phi(\mathbf{p}) : \Omega \subset \mathbb{R}^5 \rightarrow \mathbb{R}^8$. More precisely,

$$\Phi_i = \mathbf{H}_i(x, y, z) \cdot \hat{n}(\theta, \phi) \quad (20)$$

where $\mathbf{H}_i$ is the magnetic field calculated at (x, y, z) due to coil i using equation (15), and $\hat{n}(\theta, \phi)$ is the vector normal to the sensor, defined by:

$$\hat{n}(\theta, \phi) = \begin{bmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{bmatrix} \quad (21)$$

Since this maps from a subset of $\mathbb{R}^5$ to $\mathbb{R}^8$ it is clear that the forward map is not surjective.

0.3.2 The Jacobian

Now that we have established that Φ is not surjective we must move our attention to thinking about local invertibility rather than global. To do so we define the Jacobian:

$$J(\mathbf{p}) = \frac{\partial \Phi}{\partial \mathbf{p}} \in \mathbb{R}^{8 \times 5}, J_{i,j} = \frac{\partial \Phi_i}{\partial p_j} \quad (22)$$

Since $\mathbf{H}$ is a finite sum of Biot Savart applications it is a smooth map away from the current filaments. $\hat{n}$ is also a smooth map, therefore Φ will also be

smooth away from the current filaments and the Jacobian will be well defined there.

A closed form for $\mathbf{J}$ is cumbersome find so we instead use finite differences to approximate $\mathbf{J}$

$$\mathbf{J}_{:,i} \approx \frac{\Phi(\mathbf{p} + h * \hat{e}_i) - \Phi(\mathbf{p})}{h} \quad (23)$$

where $\hat{e}_i$ is the unit vector in the i th direction, and h is a small real value ($h = 10^{-5}$ in our simulations).

0.3.3 Local Invertibility

As a result of the inverse function theorem [?, Thm 9.24]; where $\mathbf{J}(\mathbf{p})$ has full column rank, or equivalently, where $\mathbf{J}^\top \mathbf{J}$ is invertible Φ is locally injective and has a C^1 local inverse in a neighbourhood of $\Phi(\mathbf{p})$.

Rank deficiency therefore signals regions where the pose cannot be recovered from the flux, even locally. Full column rank is therefore a condition for the inverse problem to be well posed.